

The Network Origins of Inflation Stances in a Currency Union

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August 19, 2026

Research Questions

- **Questions**

- What structural forces shape cross-country divergence in inflation preferences in a currency union?
- What's the welfare implication of this divergence?

- **This paper:** proposes a production-network explanation.

- Studies optimal policy in a heterogeneous-agent input-output (HAIO) economy.
- Defines *unilateral inflation stance* and *policy-alignment loss* (PAL).
- Quantitatively: a country's inflation stance operates mainly through its *input-output multiplier*, and hence, relates to its position in the union's network.

Related Literature

- New Open Economy Macroeconomics (NOEM) literature on **currency-union policy design**: e.g., Benigno, 2004; Galí, 2008; Ferrero, 2009; Farhi and Werning, 2017.
- HANK literature on the **redistribution motive of monetary policy**: e.g., Bhandari et al., 2021; Acharya et al., 2023; Dávila and Schaab, 2023; Le Grand et al., 2022; La'O and Morrison, 2024; Nuño and Thomas, 2022.
- Production network literature on **optimal monetary policy**: e.g. La'O and Tahbaz-Salehi, 2022; Rubbo, 2023; Xu and Yu, 2025; Qiu et al., 2026.

Environment

- The framework of this paper sits at the intersection of Baqaee and Farhi (2024) and La'O and Tahbaz-Salehi (2022).
- Static New Keynesian economy with C countries and N industries.
- Each country has a representative household that supplies a distinct type of labor.
- Countries differ in labor supply elasticity, sectoral labor input shares, ownership structures, and consumption baskets.
- The general formulation does not require geographic segmentation; the calibration imposes it (each country associated with a subset of industries N_c).

Firms

- A unit mass of monopolistically competitive firms $k \in [0, 1]$ in sector i

$$y_{ik} = z_i F_i(\{x_{ij,k}\}_{j=1}^N, \{L_{ic}\}_{c=1}^C) = z_i \varsigma_i \prod_{j=1}^N x_{ij,k}^{\omega_{ij}} \prod_{c=1}^C L_{ic,k}^{\alpha_{ic}}$$

- In the calibration (geographic segmentation), α_{ic} equals the value-added share of industry i if it is located in country c and zero otherwise.
- Differentiated varieties within an industry i aggregated via a CES aggregator

$$y_i = \left(\int_0^1 y_{ik}^{\frac{\theta_i-1}{\theta_i}} dk \right)^{\frac{\theta_i}{\theta_i-1}}$$

- All firms in industry i solve the following cost-minimization problem:

$$mc_i = \min_{\{x_{ij,k}\}, \{L_{ic,k}\}} \sum_{c=1}^C w_c L_{ic,k} + \sum_{j=1}^N p_j x_{ij,k}, \quad s.t. \ y_{ik} = 1$$

Nominal Rigidities

- Firm k in industry i can reset price with probability $\delta_i \in (0, 1]$. The optimal reset price maximizes the profit

$$\Pi_{ik} = \max_{p_{ik}} [p_{ik} - (1 - \tau_i)mc_i] \left(\frac{p_{ik}}{p_i} \right)^{-\theta_i} y_i$$

- To a first-order approximation,

$$\log \mathbf{p} = \delta \log \mathbf{mc}, \quad \log \boldsymbol{\mu} = -(\delta^{-1} - I) \log \mathbf{p}$$

where $\delta = \text{diag}(\delta_1, \dots, \delta_N)$.

Households

- Utility function of household c

$$U_c(C_c, L_c) = \log(C_c) - \psi_c \frac{L_c^{1+1/\eta_c}}{1 + 1/\eta_c}$$

where C_c is a utility-relevant aggregator of sectoral goods consumed by household c

$$C_c = \prod_{i=1}^N (c_{ci}/\beta_{ci})^{\beta_{ci}}.$$

- Budget constraint

$$\text{GNE}_c \equiv \sum_{i=1}^N p_i c_{ci} \leq w_c L_c + \sum_{i=1}^N \Phi_{ic} \Pi_i - T_c$$

where Φ is the ownership matrix (geographic segmentation: $\Phi_{ic} = \mathbf{1}\{i \in N_c\}$).

Policy Instruments

- **Fiscal policy:** industry subsidies $1 - \tau_i = (\theta_i - 1)/\theta_i$ eliminate CES distortions; financed by lump-sum taxes proportional to ownership shares,

$$T_c = \sum_{i \in N} \Phi_{ic} \tau_i m c_i \int_0^1 y_{ik} dk.$$

- **Monetary policy:** controls nominal aggregate demand

$$\text{GNE} = \sum_{c=1}^C \sum_{i=1}^N p_i c_{ci} = m.$$

Equilibrium Definition

Definition

A *sticky-price equilibrium* is a set of allocations, prices, and policies such that:

- (i) firms optimally choose inputs and reset prices when allowed;
- (ii) each household maximizes utility subject to its budget constraint;
- (iii) the government budget constraint is satisfied;
- (iv) all markets clear.

Definition

The *flexible-price equilibrium* is defined by the same conditions, except that all firms reset their prices optimally.

- Log deviation from the flexible-price equilibrium: $\hat{x} = \log x - \log x^*$.

Minimum Notations

Table: Country-level and union-level notation

	Country c	Union aggregate
Income (GNE) share	$\chi_c = \frac{\text{GNE}_c}{\text{GNE}}$	$\sum_{c=1}^C \chi_c = 1$
Consumption share	β_{ci}	$b_i = \sum_{c=1}^C \chi_c \beta_{ci}$
Domar weight of sector i	$\lambda_i^c = \sum_j \beta_{cj} \Psi_{ji}$	$\lambda_i = \sum_{c=1}^C \chi_c \lambda_i^c$

- β_{ci} is direct exposure of country c 's household to sector i .
- λ_i^c is direct and indirect exposure of country c 's household to sector i .

Allocative Efficiency

Proposition

To a first-order approximation, the country level allocative efficiency is

$$\hat{c}_c - \hat{l}_c = \underbrace{\sum_{i=1}^N \left(\frac{\lambda_i \Phi_{ic}}{\chi_c} - \lambda_i^c \right) \log \mu_i}_{\text{direct-incidence (DI)} \equiv \mathcal{J}_c^{\text{DI}'} \log \mu} + \underbrace{d \log w_c - \sum_{f=1}^C \Lambda_f^c d \log w_f}_{\text{factoral terms-of-trade (FToT)} \equiv \mathcal{J}_c^{\text{FToT}'} \log \mu} \equiv \mathcal{J}_c' \log \mu$$

where FToT component can be rewritten in terms of sectoral markups as

$$\mathcal{J}_c^{\text{FToT}'} \log \mu = \sum_{i=1}^N \left[\left(\Gamma_{ic} + \frac{1}{\eta_c} \ell_{ic}^\mu \right) - \sum_{f=1}^C \Lambda_f^c \left(\Gamma_{if} + \frac{1}{\eta_f} \ell_{if}^\mu \right) \right] \log \mu_i$$

with Γ and ℓ^μ denoting Jacobian matrices

$$\Gamma_{ic} \doteq \frac{d \log \chi_c}{d \log \mu_i} = \chi_c^{-1} \lambda_i \sum_{f=1}^C Q_{cf} (\Phi_{if} - \Psi_{if}), \quad \ell_{ic}^\mu \doteq \frac{d \log L_c}{d \log \mu_i} = -\frac{1}{1 + 1/\eta_c} \cdot \frac{\Phi_{ic} \lambda_i}{\chi_c}.$$

Allocative Efficiency as Overall Terms of Trade

- In a horizontal economy (without input–output linkages), and assuming ownership segmentation

$$\mathcal{J}_c^{\text{DI}'} \log \mu = \underbrace{\sum_{i \in N_c} \sum_{f \neq c} \chi_f \beta_{fi} / \chi_c \log \mu_i}_{\text{change in export price}} - \underbrace{\sum_{i \notin N_c} \beta_{ci} \log \mu_i}_{\text{change in import price}} .$$

- The direct incidence effect maps to the commodity terms of trade effect.
- Allocative efficiency = overall (commodity + factoral) terms of trade effect

The Role of Production Networks

- Under ownership segmentation, direct-incidence effect can be further decomposed into:

$$\mathcal{J}_c^{\text{DI}} \log \mu = \underbrace{\left(\xi_c - \mathbb{E}_{\chi}[\xi] \right) \overline{\log \mu}}_{\text{IO-multiplier channel}} + \underbrace{\left(\mathbb{E}_{\chi}[\xi] - \sum_{i \in N} \lambda_i^c \right) \overline{\log \mu}}_{\text{home-bias channel}} + \underbrace{\sum_{i \in N} \mathcal{J}_{ic}^{\text{DI}} (\log \mu_i - \overline{\log \mu})}_{\text{sectoral-heterogeneity channel}}.$$

- Shutting down production networks loses first two channels ($\xi_c = 1$ for all c).
- When the IO-multiplier channel dominates, countries with above average ξ_c experience a deterioration in their terms of trade under monetary expansion (markup compression).

Definition (Input–Output Multiplier)

The *input–output multiplier* of country c is defined as

$$\xi_c \equiv \sum_{i \in N_c} \frac{\lambda_i}{\chi_c} = \frac{\sum_{i \in N_c} p_i y_i}{\text{GNE}_c},$$

which measures total domestic sales as a fraction of domestic final demand. The union-wide counterpart is $\mathbb{E}_{\chi}[\xi] = \sum_{i \in N} \lambda_i$.

Aggregate Cancellation: A Macro Envelope Theorem

Corollary

Country-level allocative efficiencies, while generally nonzero, exactly offset at first order when aggregated using income shares:

$$\sum_{c=1}^C \chi_c (\hat{c}_c - \hat{l}_c) = \sum_{c=1}^C \chi_c \mathcal{J}'_c \log \mu = 0.$$

- Intuition: zero terms of trade in a closed economy.
- Implication: monetary policy carries a redistribution motive whenever welfare aggregation departs from income-share weighting.

Bergson–Samuelson Welfare Aggregation

- Adopt a Bergson–Samuelson aggregator

$$W(\{\kappa_c\}_{c=1}^C) = \sum_{c=1}^C \kappa_c U_c,$$

where $\{\kappa_c\}$ are Pareto weights satisfying $\sum_c \kappa_c = 1$.

- Unilateral weighting $\mathbf{e}_c$ assigns $\kappa_c = 1$ and $\kappa_{c'} = 0$ for $c' \neq c$, so

$$W(\mathbf{e}_c) = U_c.$$

- This polar case isolates country c 's perspective; the general case follows by aggregating with arbitrary $\{\kappa_c\}$.

Unilateral Welfare Loss

Lemma (Unilateral Welfare Loss)

Up to a second-order approximation, the welfare loss under unilateral weights for country c is

$$W(\mathbf{e}_c) - W^*(\mathbf{e}_c) = \mathcal{J}'_c \log \boldsymbol{\mu} - \frac{1}{2} \log \boldsymbol{\mu}' \mathcal{L}_c \log \boldsymbol{\mu},$$

where $\mathcal{L}_c = \mathcal{L}_c^{\text{e.g.}} + \mathcal{L}_c^{\text{within}} + \mathcal{L}_c^{\text{across}}$ summarizes three sources of misallocation:

$$\mathcal{L}_c^{\text{e.g.}} = \frac{\eta_c}{1+\eta_c} \chi_c^{-2} [\lambda \circ \Phi_{(:,c)}][\lambda \circ \Phi_{(:,c)}]',$$

$$\mathcal{L}_c^{\text{within}} = \text{diag}(\lambda^c \circ \theta)(\delta^{-1} - I)^{-1},$$

$$\mathcal{L}_c^{\text{across}} = \frac{\eta_c}{1+\eta_c} \Upsilon^c + \sum_{f=1}^C \frac{1}{1+\eta_f} \Lambda_f^c \Upsilon^f - \Xi^c + \sum_{f=1}^C \Lambda_f^c \Xi^f.$$

where Ξ_{ij}^c and Υ_{ij}^c are income-share and labor-wedge Hessians of country c

$$\Xi_{ij}^c \equiv \frac{\partial^2 \log \chi_c}{\partial \log \mu_i \partial \log \mu_j} = -\Gamma_{jc} \Gamma_{ic} + \chi_c^{-1} \sum_{f=1}^C Q_{cf} (\Phi_{if} - \Psi_{if}) \left(2 \sum_{g=1}^C \lambda_i^g \chi_g \Gamma_{jg} + \lambda_i t_{ij} - 2 \lambda_j \Psi_{ji} \right),$$

$$\Upsilon_{ij}^c \equiv \frac{\partial^2 \log (\Lambda_c / \chi_c)}{\partial \log \mu_i \partial \log \mu_j} = \frac{\Phi_{ic}}{\chi_c} \left(2 \lambda_j \Psi_{ji} - 2 \sum_{f=1}^C \lambda_i^f \chi_f \Gamma_{jf} + 2 \lambda_i \Gamma_{jc} - \lambda_i t_{ij} - \lambda_i \lambda_j \frac{\Phi_{jc}}{\chi_c} \right).$$

Bergson–Samuelson Welfare Loss

Proposition (Bergson–Samuelson Welfare Loss)

The Bergson–Samuelson welfare loss with arbitrary weights $\{\kappa_c\}_{c=1}^C$ is the weighted sum:

$$\begin{aligned} W(\{\kappa_c\}) - W^*(\{\kappa_c\}) &= \sum_{c=1}^C \kappa_c [W(\mathbf{e}_c) - W^*(\mathbf{e}_c)] \\ &= \sum_{c=1}^C \kappa_c \mathcal{J}'_c \log \boldsymbol{\mu} - \frac{1}{2} \log \boldsymbol{\mu}' \mathcal{L}(\{\kappa_c\}) \log \boldsymbol{\mu}. \end{aligned}$$

The aggregate loss matrix is the welfare-weighted average,

$$\mathcal{L}(\{\kappa_c\}) \equiv \sum_{c=1}^C \kappa_c \mathcal{L}_c = \mathcal{L}^{\text{e.g.}}(\{\kappa_c\}) + \mathcal{L}^{\text{within}}(\{\kappa_c\}) + \mathcal{L}^{\text{across}}(\{\kappa_c\}).$$

- Generalizes Baqaee and Farhi (2024) (income-share weights) and La'O and Tahbaz-Salehi (2022) (representative agent) to a HAI0 economy with arbitrary Pareto weights.

Optimal Policy

Proposition (Optimal Monetary Policy (Second Best))

Given Pareto weights $\{\kappa_c\}_{c=1}^C$, the optimal monetary policy targets a sectoral price index

$$\sum_{j=1}^N \zeta_j^*(\{\kappa_c\}) \log p_j = \pi^*(\{\kappa_c\}), \quad \pi^*(\{\kappa_c\}) = \sum_{c=1}^C \kappa_c \mathcal{J}'_c(I - \delta^{-1}) \varrho^m,$$

where $\varrho^m \equiv \frac{d \log p}{d \log m} = [I - \varrho^w(\Gamma + \ell^u \eta^{-1})'(I - \delta^{-1})]^{-1} \varrho^w \mathbf{1}$.

The optimal sectoral weights decompose as $\zeta_j^*(\{\kappa_c\}) = \zeta_j^{\text{e.g.}} + \zeta_j^{\text{within}} + \zeta_j^{\text{across}}$, with

$$\zeta_j^{\text{e.g.}} = (\delta_j^{-1} - 1) \lambda_j \sum_{c=1}^C \kappa_c \Phi_{jc} \ell_c^m / \chi_c,$$

$$\zeta_j^{\text{within}} = (\delta_j^{-1} - 1) \theta_j \varrho_j^m \sum_{c=1}^C \kappa_c \lambda_j^c,$$

$$\zeta_j^{\text{across}} = (\delta_j^{-1} - 1) \sum_{i=1}^N \sum_{c=1}^C \kappa_c (\delta_i^{-1} - 1) \varrho_i^m \mathcal{L}_{c,ij}^{\text{across}}.$$

Centralized Optimal Policy vs. Unilateral Optimal Policy

- **Centralized optimal monetary policy** ($\kappa_c = \chi_c$): pure price-stabilization rule with zero inflation bias, $\sum_{j=1}^N \zeta_j^*(\{\chi_c\}) \log p_j = 0$.
- **Unilateral optimal policy** ($\{\kappa_c\} = \mathbf{e}_c$): minimizes country c 's welfare loss, $\sum_{j=1}^N \zeta_j^*(\mathbf{e}_c) \log p_j = \pi^*(\mathbf{e}_c)$.
- Not straightforward to compare unilateral optimal policies across countries: each country has its own preferred price index and inflation bias.

Unilateral Inflation Stance

- Take the union's target price index ζ as given.

Definition (Unilateral Inflation Stance (Third Best))

The unilateral inflation stance of country c is the inflation rate it selects to minimize its expected welfare loss, taking the union's regime ζ as given:

$$\pi_c(\zeta) = \arg \min_{\pi \in \mathbb{R}} \mathbb{E}[\mathbb{L}_c(\zeta, \pi)].$$

Proposition

The unilateral inflation stance satisfies

$$\pi_c(\zeta) = \underbrace{\frac{\zeta' \varrho^m}{[\zeta^*(\mathbf{e}_c)]' \varrho^m}}_{\text{price-index adjustment}} \pi^*(\mathbf{e}_c).$$

Policy-Alignment Loss (PAL)

Definition (Policy-Alignment Loss)

The *policy-alignment loss* for country c under a union-wide policy (ζ, π) is

$$\text{PAL}_c(\zeta, \pi) \equiv \mathbb{L}_c(\zeta, \pi) - \mathbb{L}_c(\zeta^*(\mathbf{e}_c), \pi^*(\mathbf{e}_c)) \geq 0.$$

Proposition (PAL Decomposition)

The expected PAL admits the closed-form decomposition

$$\mathbb{E}[\text{PAL}_c(\zeta, \pi)] = \underbrace{\frac{1}{2} \frac{\zeta^*(\mathbf{e}_c)' \varrho^m}{(\zeta' \varrho^m)^2} [\pi_c(\zeta) - \pi]^2}_{\text{inflation misalignment}} + \underbrace{\mathcal{I}_c(\zeta)}_{\text{price-index misalignment}},$$

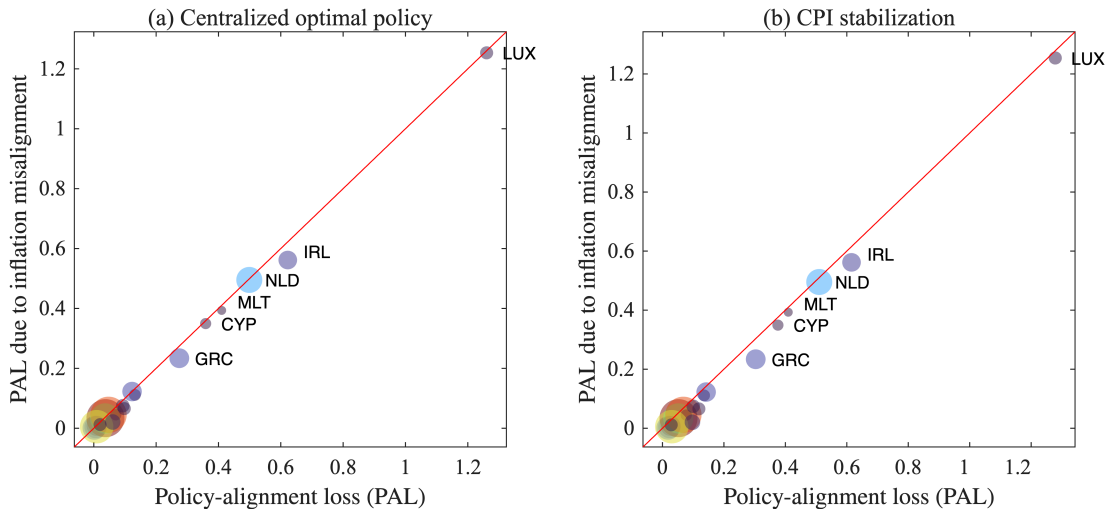
where $\mathcal{I}_c(\zeta) \geq 0$ vanishes at $\zeta = \zeta^*(\mathbf{e}_c)$.

- Inflation misalignment is **quadratic** in the distance $|\pi_c(\zeta) - \pi|$.

Calibration

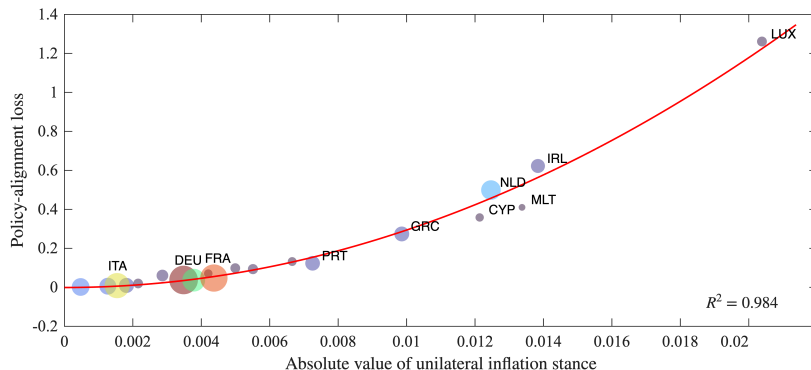
- **Parameters:** $\eta_c = 2$, $\theta_i = 6$, $\Phi_{ic} = \mathbf{1}\{i \in N_c\}$ (geographically segmented ownership).
- **IO matrix:** WIOD 2014 (Timmer et al., 2015); $C = 20$ euro-area countries, each with $N_c = 56$ industries.
- **Price rigidities:** monthly FPA from Pastén et al. (2020), mapped to WIOD industries via NAICS; converted to quarterly.
- **Sectoral productivity shocks:** annual TFP growth from WIOD SEA via Törnqvist index, interpolated to quarterly, linearly detrended, and used to form the variance–covariance matrix.

Decomposition of Policy-Alignment Loss



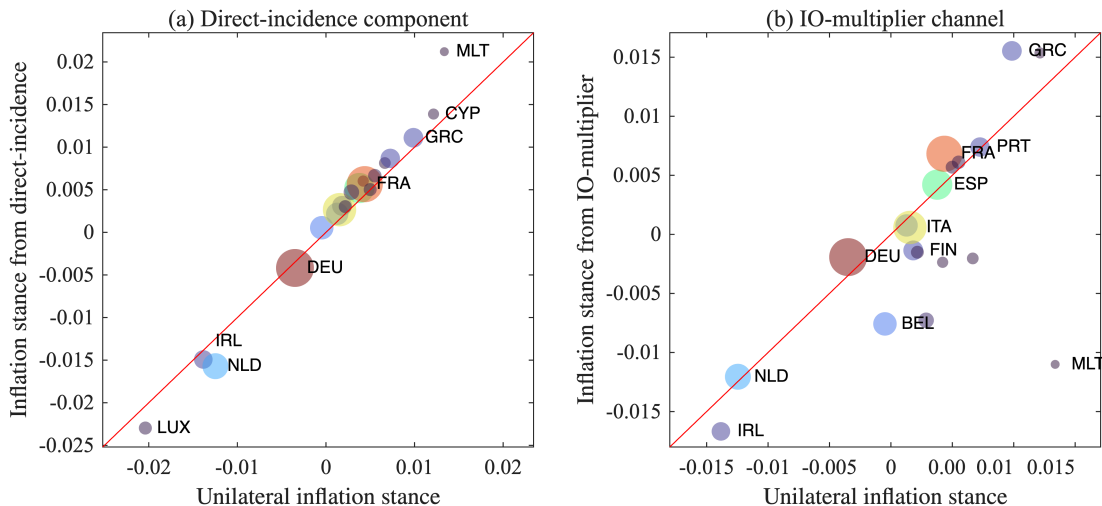
- PAL is mainly driven by inflation misalignment (quadratic in inflation stance deviation).

Inflation-Stance Deviation and PAL: Quadratic Fit



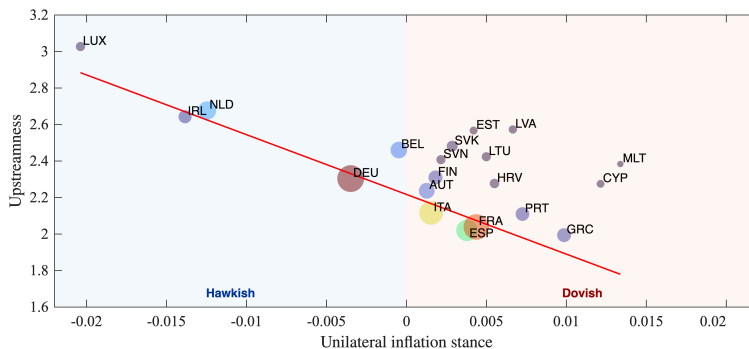
- Union-wide inflation consensus is zero under the centralized optimal policy.
- Quadratic fit (no linear term) gives $R^2 = 0.98$.
- A one-percentage-point inflation deviation predicts a loss of about 0.29 pp of steady-state consumption.
- Tail countries (LUX, IRL, NLD, GRC) bear disproportionate welfare costs.

Decomposition of the Inflation Stance



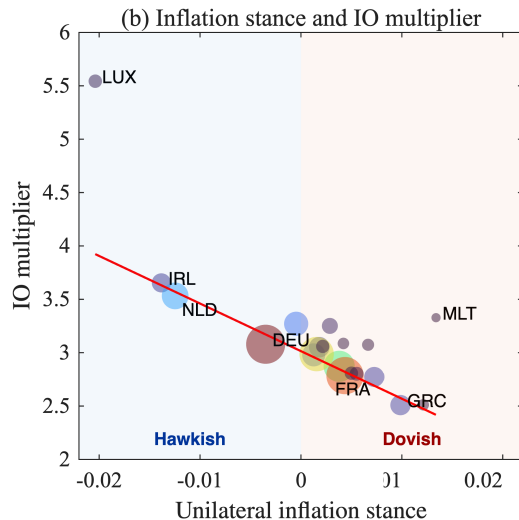
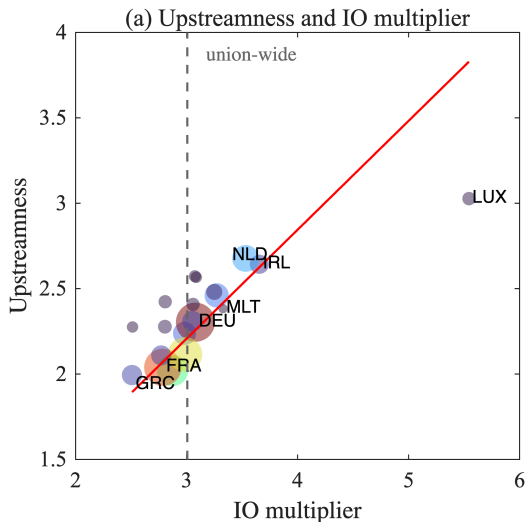
- Panel (a): The direct-incidence component reproduces almost the entire unilateral inflation stance.
- Panel (b): the *IO-multiplier channel* alone reproduces most cross-country variation.

Inflation Stance and Upstreamness



- **Upstreamness** (Antràs et al., 2012): value-added-weighted distance from final consumption.
- Strong negative relationship: WLS slope ≈ -32.7 , $R^2 = 0.77$.
- More upstream countries prefer lower inflation; downstream prefer higher.
- Mirrors the familiar North–South divide in the euro area.

IO Multiplier as Mediating Network Statistic



- Panel (a): upstreamness $\leftrightarrow$ local IO multiplier ($R^2 = 0.77$).
- Panel (b): IO multiplier $\leftrightarrow$ unilateral inflation stance ($R^2 = 0.75$).
- Mechanism: network position $\Rightarrow$ IO multiplier $\Rightarrow$ change in terms of trade $\Rightarrow$ inflation preference.

Conclusion

- This paper develops a HAIO model and provides microfoundations for divergent inflation preferences across member states in a currency union.
- Bringing the model to the euro area, I find that a country's inflation stance negatively relates to its upstreamness, and the deviation from the union-wide consensus predicts welfare costs. Therefore, countries at either end of the production chain exhibit extreme inflation stances and bear the largest welfare costs.

References

- Acharya, Sushant, Edouard Challe, and Keshav Dogra (2023) 'Optimal monetary policy according to HANK.' *American Economic Review* 113(7), 1741–1782
- Antràs, Pol, Davin Chor, Thibault Fally, and Russell Hillberry (2012) 'Measuring the upstreamness of production and trade flows.' *American Economic Review* 102(3), 412–416
- Baqaei, David R., and Emmanuel Farhi (2024) 'Networks, Barriers, and Trade.' *Econometrica* 92(2), 505–541
- Benigno, Pierpaolo (2004) 'Optimal monetary policy in a currency area.' *Journal of International Economics* 63(2), 293–320
- Bhandari, Anmol, David Evans, Mikhail Golosov, and Thomas J Sargent (2021) 'Inequality, Business Cycles, and Monetary-Fiscal Policy.' *Econometrica* 89(6), 2559–2599
- Dávila, Eduardo, and Andreas Schaab (2023) 'Optimal monetary policy with heterogeneous agents: Discretion, commitment, and timeless policy.' Technical Report, National Bureau of Economic Research
- Farhi, Emmanuel, and Iván Werning (2017) 'Fiscal unions.' *American Economic Review* 107(12), 3788–3834
- Ferrero, Andrea (2009) 'Fiscal and monetary rules for a currency union.' *Journal of International Economics* 77(1), 1–10
- Galí, Jordi (2008) *Monetary policy, inflation, and the business cycle: an introduction to the new Keynesian framework* (Princeton University Press)
- La'O, Jennifer, and Alireza Tahbaz-Salehi (2022) 'Optimal monetary policy in production networks.' *Econometrica* 90(3), 1295–1336
- La'O, Jennifer, and Wendy A Morrison (2024) 'Optimal monetary policy with redistribution.' Technical Report, National Bureau of Economic Research
- Le Grand, François, Alaïs Martin-Baillon, and Xavier Ragot (2022) 'Should monetary policy care about redistribution? Optimal monetary and fiscal policy with heterogeneous agents'
- Nuño, Galo, and Carlos Thomas (2022) 'Optimal redistributive inflation.' *Annals of Economics and Statistics* (146), 3–64
- Pastén, Ernesto, Raphael Schoenle, and Michael Weber (2020) 'The propagation of monetary policy shocks in a heterogeneous production economy.' *Journal of Monetary Economics* 116(C), 1–22
- Qiu, Zhesheng, Yicheng Wang, Le Xu, and Francesco Zanetti (2026) 'Monetary policy in open economies with production networks.' *Journal of Monetary Economics* 159, 103918
- Rubbo, Elisa (2023) 'Networks, phillips curves, and monetary policy.' *Econometrica* 91(4), 1417–1455
- Timmer, Marcel P., Erik Dietzenbacher, Bart Los, Robert Stehrer, and Gaaitzen J. Vries (2015) 'An illustrated user guide to the world input–output database: The case of global automotive production.' *Review of International Economics* 23(3), 575–605
- Xu, Zhihao, and Changhua Yu (2025) 'Optimal monetary policy in production networks with distortions.' *Journal of Economic Theory* 225, 105979